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Report

# Imp Promotes Axonal Remodeling by Regulating *profilin* mRNA during Brain Development

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## Summary

**Neuronal remodeling is essential for the refinement of neuronal circuits in response to developmental cues [1–4]. Although this process involves pruning or retraction of axonal projections followed by axonal regrowth and branching, how these steps are controlled is poorly understood. *Drosophila* mushroom body (MB) γ neurons provide a paradigm for the study of neuronal remodeling, as their larval axonal branches are pruned during metamorphosis and re-extend to form adult-specific branches [5]. Here, we identify the RNA binding protein Imp as a key regulator of axonal remodeling. Imp is the sole fly member of a conserved family of proteins that bind target mRNAs to promote their subcellular targeting [6–12]. We show that whereas Imp is dispensable for the initial growth of MB γ neuron axons, it is required for the regrowth and ramification of axonal branches that have undergone pruning. Furthermore, Imp is actively transported to axons undergoing developmental remodeling. Finally, we demonstrate that *profilin* mRNA is a direct and functional target of Imp that localizes to axons and controls axonal regrowth. Our study reveals that mRNA localization machineries are actively recruited to axons upon remodeling and suggests a role of mRNA transport in developmentally programmed rewiring of neuronal circuits during brain maturation.**

## Results and Discussion

### Imp Localizes to the Axons of Mushroom Body γ Neurons from Metamorphosis Onward

In cultured vertebrate neurons, ZBP1 mediates the transport of *β-actin* mRNA to axons, a process required for the chemiotropic response of growth cones to guidance cues [8, 10, 11]. Whether these observations reflect a general requirement for ZBP1 and axonal mRNA transport during brain development has remained unclear [13, 14]. We found that Imp, the *Drosophila* ZBP1 ortholog, accumulated in the cell bodies of a large number of neural cells in adult brain (Figures S1A and S1B available online) [15, 16]. Strikingly, Imp was additionally observed in the axonal compartment of a subpopulation of mushroom body (MB) neurons. MBs are composed of three main neuronal types (αβ, α′β′, and γ) with specific axonal projection patterns and developmental programs (Figures 1A and 1B) [5]. α′β′ and αβ neurons are generated during late larval stage and early metamorphosis and are maintained until adulthood. γ neurons are born during late embryogenesis and early larval stages and undergo extensive remodeling during metamorphosis [5, 17]. As shown in Figures 1C, S1C, and S1D, Imp was found to be enriched in adult γ neuron axons where it colocalized with FasciclinII, but it was not detected in the axons of nonremodeling MB neurons (αβ and α′β′ neurons). To test whether Imp is expressed in αβ and α′β′ neurons, we labeled brains expressing GFP in γ and αβ-core neurons with antibodies against Imp and Trio, a protein specifically expressed in adult α′β′ and γ neurons [18]. Imp was not detected in αβ-core neurons but accumulated in the cell bodies of both α′β′ and γ neurons (Figure 1D). Thus, both the expression and subcellular distribution of Imp are tightly regulated in *Drosophila* MB neurons (Figure S1E).

To investigate whether Imp axonal translocation is developmentally regulated, we analyzed the distribution of Imp within γ neurons at different stages. In third-instar larvae, Imp accumulated exclusively in the cell bodies and was not observed in axons (Figures 1E and S1G). During metamorphosis (pupariation), MB γ neurons first prune the distal part of their axons and then re-extend a medial branch to establish adult-specific projections (Figure S1F) [17]. Six hours after puparium formation (APF), Imp was weakly detected in γ neuron axons (Figure S1G). Such an axonal accumulation of Imp was visible at the time larval γ neurons have completed the pruning of their axonal processes (18 hr APF). During the subsequent intensive growth phase, Imp was enriched at the tip of axons, where it accumulated in particles (inset in Figure S1G; 26 hr APF). Thus, the translocation of Imp to axons is developmentally controlled, and correlates with axonal remodeling.

### Imp Is Required for the Regrowth and Branching of Pruned γ Axons

To test whether Imp is required for γ axon developmental remodeling, we analyzed the morphology of adult homozygous mutant neurons generated using the MARCM (mosaic analysis with a repressible cell marker) system [19]. Clones in which the entire progeny of a neuroblast was mutant exhibited a reduced number of cells and an altered morphology (Figures S2A and S2B). Although wild-type adult γ axons typically span the entire medial lobe, a mixture of elongated and nonelongated axons was observed upon *imp* inactivation (Figure S2B). To better visualize the morphology of mutant neurons, we analyzed single labeled neurons. Wild-type adult γ neurons extend one main axonal process that reaches the extremity of the MB medial lobe (Figure 2A) [5]. Several secondary branches typically form along this main axonal process. In contrast, about 50% of *imp* γ axons failed to reach the end of the medial lobe (Figures 2B–2D). These defects did not result from axon retraction, as the proportion of defective axons did not increase with age (Figure S2C). Interestingly, mutant axons of normal length but lost directionality were observed (Figure 2C), suggesting that *imp* may be required for the response of γ axons to guidance cues during metamorphosis. *imp* mutant neurons also exhibited an overall decrease

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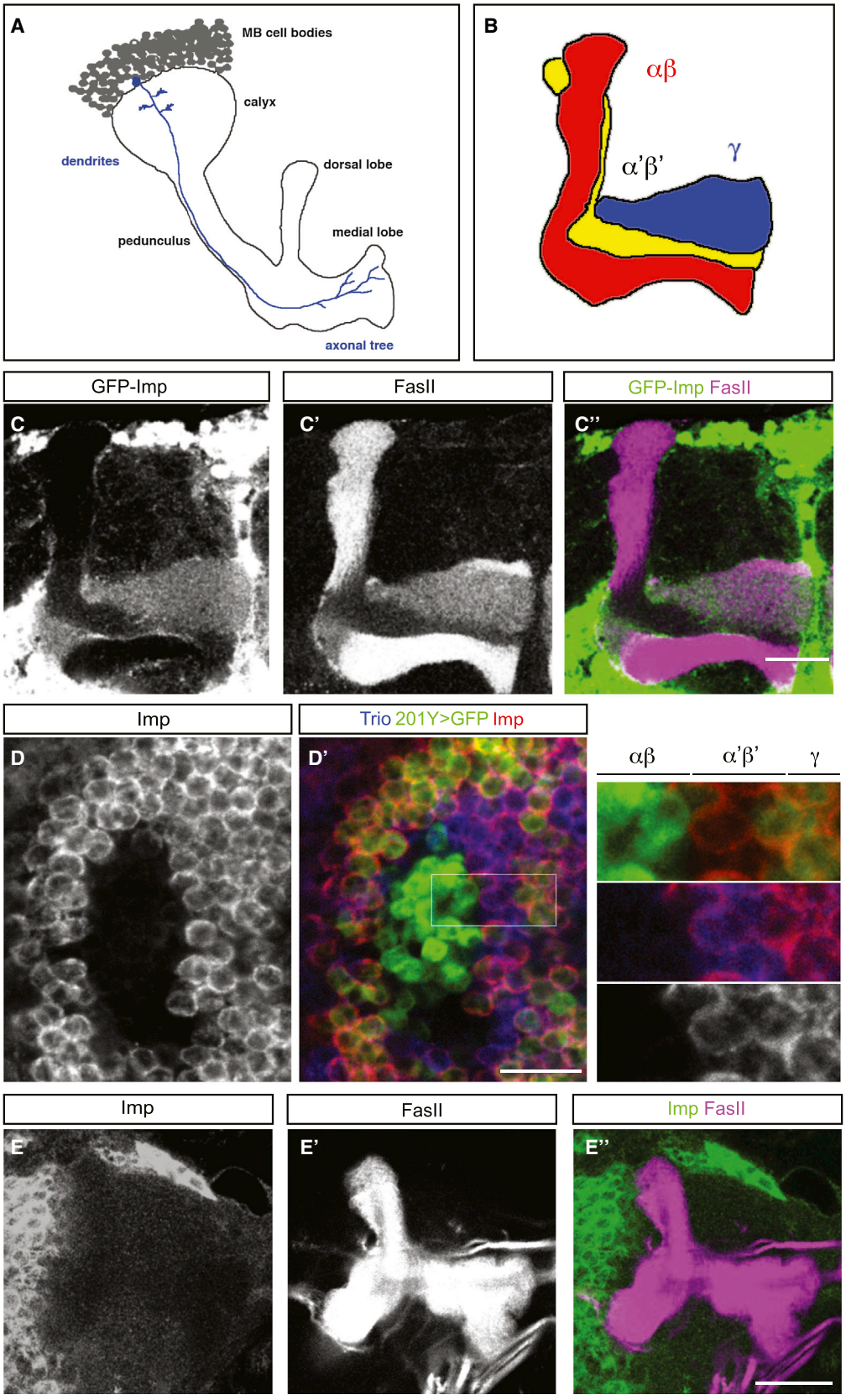



Figure 1. Imp Distribution in Mushroom Body Neurons

(A and B) Schematic representation of an adult MB (A). MB neurons (about 2,000 in total) have a stereotypic organization, with their cell bodies located dorsally at the brain periphery, their dendrites projecting just beneath in the calyx region, and their axons projecting more ventrally. Axonal processes

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in the complexity of axonal arborization patterns characterized by a reduced number of terminal branches (Figures 2E and S2D–S2F). Both phenotypes were significantly suppressed upon expression of a wild-type copy of Imp in γ neurons, revealing that *imp* acts cell autonomously to control axonal regrowth and branching (Figures 2D, 2E and S2F).

To determine whether Imp function in axonal growth correlates with its accumulation in axons, we investigated the requirement for Imp in two neuronal cell types where it is exclusively detected in cell bodies: larval γ neurons and α′β′ neurons. Both single larval γ neurons (Figures 2F–2H) and single adult α′β′ neurons (data not shown) mutant for *imp* projected their axons normally. Furthermore, larval γ neuron neuroblast clones exhibited a normal morphology, confirming that *imp* is not necessary for initial axon growth (Figures S2G and S2H). These results show that Imp is specifically required for the growth and branching of remodeling γ axons and suggest that its translocation to axons is critical for this function.

### Imp Is Transported to γ Axons as Particles Undergoing Active Motion

To address whether Imp is transported actively to the axons of regrowing γ neurons, we developed a live-imaging protocol using cultured pupal brains expressing functional GFP-Imp fusions (see rescue experiments in Figure S3A) specifically in γ neurons (201Y-Gal4, UAS-GFP-Imp). The culture conditions supported efficient axonal growth, as MB neurons from cultured brains grew similarly to their counterparts developing inside the pupa (Figure S3B). Fast confocal imaging of axon bundles (Figure 3A, left) revealed that GFP-Imp fusions accumulated in particles undergoing bidirectional movement (Movie S1). In contrast, no particles could be detected upon expression of GFP alone (Figure S3C). Motile GFP-Imp particles were distributed into three classes: particles with a strong net anterograde (56%) or retrograde (36%) movement and particles with little net bias (8%) (Figures 3A–3C; see the Supplemental Experimental Procedures). Individually tracked particle trajectories were broken into segments to calculate velocities. Segmental velocities distributed over a wide range (Figure S3D), with mean anterograde and retrograde segmental velocities of 0.98 ± 0.05 μm/s and 0.73 ± 0.03 μm/s, respectively (Figure 3D). Furthermore, curves matching the graph of a quadratic function were obtained upon plotting of the mean square displacement (MSD) values over time (Figure 3E), indicating that GFP-Imp particles undergo directed transport rather than diffusion (see the Supplemental Experimental Procedures). To assess the role of microtubules (MTs) in this process, we treated brains with colchicine. This treatment abolished MT dynamics, as revealed by the loss of EB1-GFP comets characteristic of growing MT plus ends (Figures S3E and S3F) [20]. Strikingly, motile GFP-Imp particles were no longer observed under these conditions (Figure 3F and Movie S2). These results demonstrate that Imp is a component of particles undergoing active MT-dependent transport during midpupariation, consistent with a role of Imp in the transport of selected mRNAs to regrowing γ axons.

### Imp Associates with the 3′ Untranslated Region of *chickadee* mRNA

Previous in vitro studies have revealed that the axons of immature neurons are enriched in mRNAs encoding regulators of the actin cytoskeleton that play critical roles in axonal growth and guidance [21, 22]. To identify Imp mRNA targets, we thus performed an immunoprecipitation RT-PCR-based screen for mRNAs encoding actin regulators (Table S2). We discovered that Imp selectively associates with *chickadee* (*chic*) mRNA (Figure 4A), which encodes the G-actin binding protein Profilin [23]. As revealed by affinity pull-down assays, endogenous Imp associated with the *chic* 3′ untranslated region (UTR), but not with the *chic* coding sequence (Figure 4B and S4A). To test whether Imp can interact with *chic* mRNA directly, we analyzed the binding of recombinant MBP-Imp to the *chic* 3′ UTR in electrophoretic mobility shift assays. Retarded complexes formed in the presence of the *chic* 3′ UTR, but not in the presence of a nonrelated RNA (*y14*) (Figures 4C and S4B). Furthermore, no significant interaction was observed when other MBP-tagged proteins were used (Figures S4C). Notably, two discrete complexes were detected in the presence of low amounts of Imp (Figure 4C, asterisks), whereas higher-order complexes were formed with increasing amounts of Imp (arrow). Formation of these complexes was outcompeted by the addition of nonlabeled RNAs corresponding to the *chic* 3′ UTR, but not to the *chic* coding sequence (Figure S4D). Altogether, these results show that Imp associates with *chic* mRNA in vivo and that it can bind directly and specifically to the *chic* 3′ UTR.

### *chickadee* mRNA Localizes to Growing γ Neurites

To test whether *chic* mRNA localizes to the neurites of regrowing γ neurons, we first performed in situ hybridization on pupal and adult brains. The poor signal-to-noise ratio obtained with this method at these stages, combined with the relatively low levels of axonally localized mRNAs, did not allow us to unambiguously detect *chic* transcripts or reporters in axons (data not shown). We thus used *chic* reporter constructs expressed under the control of the γ-specific 201Y-Gal4 driver and performed fluorescent in situ hybridizations on dissociated neurons extracted from 24 hr APF pupae and cultured for 3–4 days. *chic* reporter mRNAs could be observed in the neurites of γ neurons at a significantly higher frequency than control *gfp* mRNAs (Figures 4D–4F). Furthermore, *chic* mRNA and Imp colocalized in developing neurites (Figure 4G), consistent with their association within mRNA transport complexes.

To test whether the region of *chic* bound by Imp is required for *chic* mRNA localization to developing neurites, we compared the distribution of reporters containing both the *chic* coding sequence and 3′ UTR with that of reporters lacking

fasciculate in a structure called pedunculus that divides to produce a dorsal and a medial lobe. The morphology of a single neuron belonging to the γ population is shown in blue. A schematic representation of MB lobes is shown in (B).

(C–C″) Distal part of adult MB axons expressing GFP-Imp from the endogenous *imp* locus (protein-trap line PG080) and double-stained with anti-GFP (C; green in C″) and anti-FasciclinII (C′; magenta in C″) antibodies. FasciclinII (FasII) accumulates at high levels in αβ axons, accumulates at low levels in γ axons, and is not expressed in α′β′ neurons. Scale bar, 20 μm.

(D and D′) Cell bodies of a 201Y-Gal4, UAS-GFP adult brain triple stained with anti-imp (D; red in D′), anti-Trio (blue in D′), and anti-GFP (green in D′) antibodies. Scale bar, 10 μm. Right: close-up views of the region boxed in (D′). The lower panel corresponds to anti-Imp staining.

(E–E″) Distal part of third instar larval MB axons double-stained with anti-Imp (E; green in E″) and anti-FasII (E′; magenta in E″) antibodies. Scale bar, 20 μm. See also Figure S1.

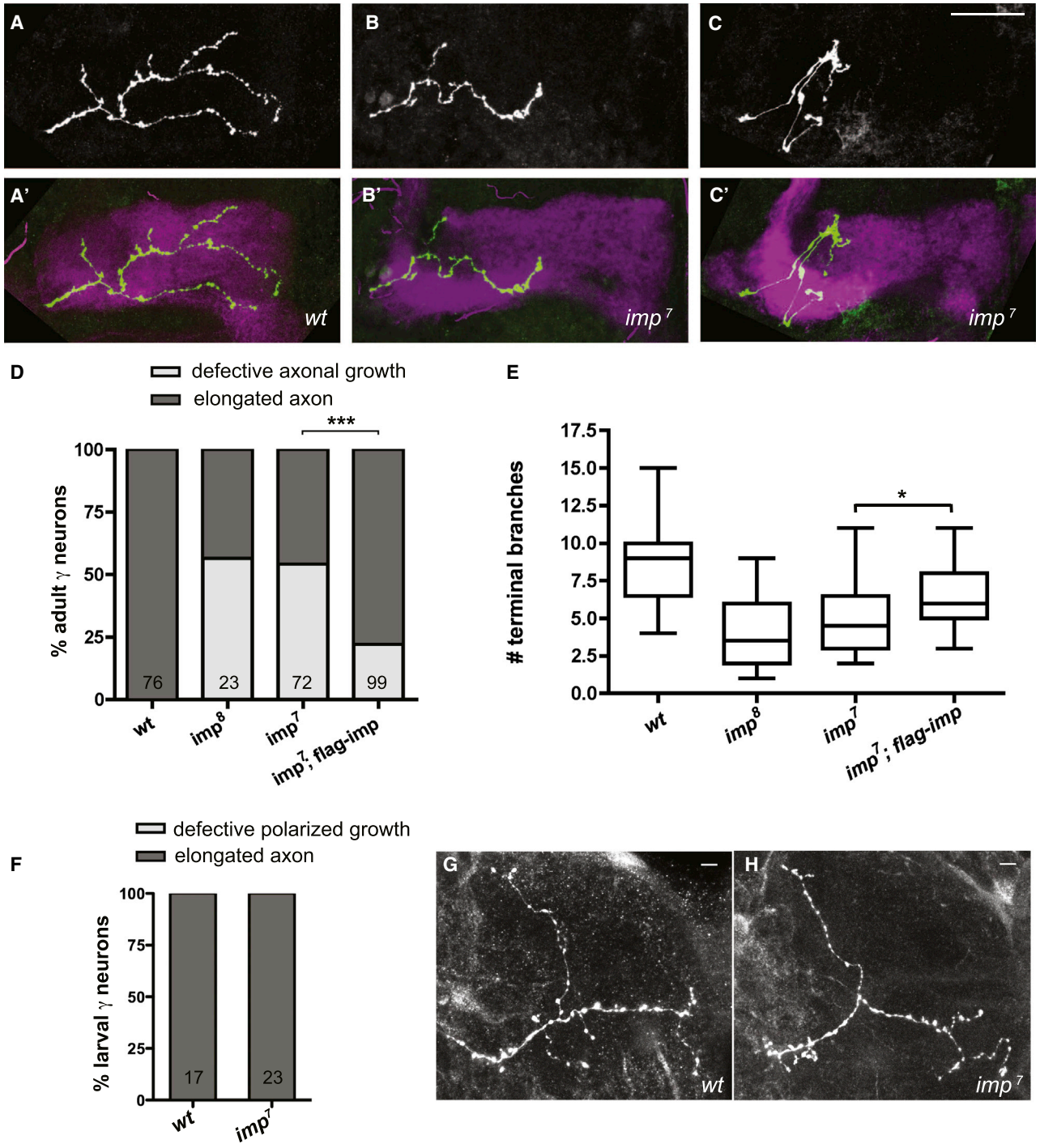



Figure 2. Axonal Growth and Branching Defects in γ Neurons Mutant for *imp*

(A–C′) Axonal projections of single wild-type (A) and *imp*$^{7}$ (B and C) adult γ neurons labeled by GFP (A–C; green in A′–C′) using the MARCM technique. The shape of the MB medial lobe is highlighted by FasII staining (magenta in A′–C′). Scale bar, 20 μm.

(D) Percentages of single adult γ axons that succeeded (elongated axon) or failed (defective axonal growth) to reach the extremity of the medial lobe. *imp*$^{7}$*; flag-imp* corresponds to the *imp* rescue context. ***p < 0.001 (Fisher's exact test). Numbers correspond to the total numbers of scored axons.

(E) Number of terminal axonal branches per adult γ neuron. Numbers of scored axons are as follows: 11 (WT), 10 (*imp*$^{8}$), 24 (*imp*$^{7}$), and 15 (*imp*$^{7}$*; flag-imp*). *p < 0.05 (Kruskal-Wallis test with Dunn's post test).

(F) Percentages of single larval γ neurons with axonal growth defects.

(G and H) Axonal projections of single wild-type (G) and *imp*$^{7}$ (H) larval γ neurons labeled with GFP. Scale bar, 5 μm.

Complete genotypes are as follows: FRT19A, tub-Gal80, hsp-flp/FRT19A; 201Y-Gal4, UAS-GFP/+ (WT); FRT19A, tub-Gal80, hsp-flp/FRT19A *imp*$^{7\ or\ 8}$; 201Y-Gal4, UAS-GFP/+ (*imp* mutants) and FRT19A, tub-Gal80, hsp-flp/FRT19A *imp*$^{7}$; 201Y-Gal4, UAS-GFP/+; UAS-flag-*imp*/+ (rescue condition). See also Figure S2.

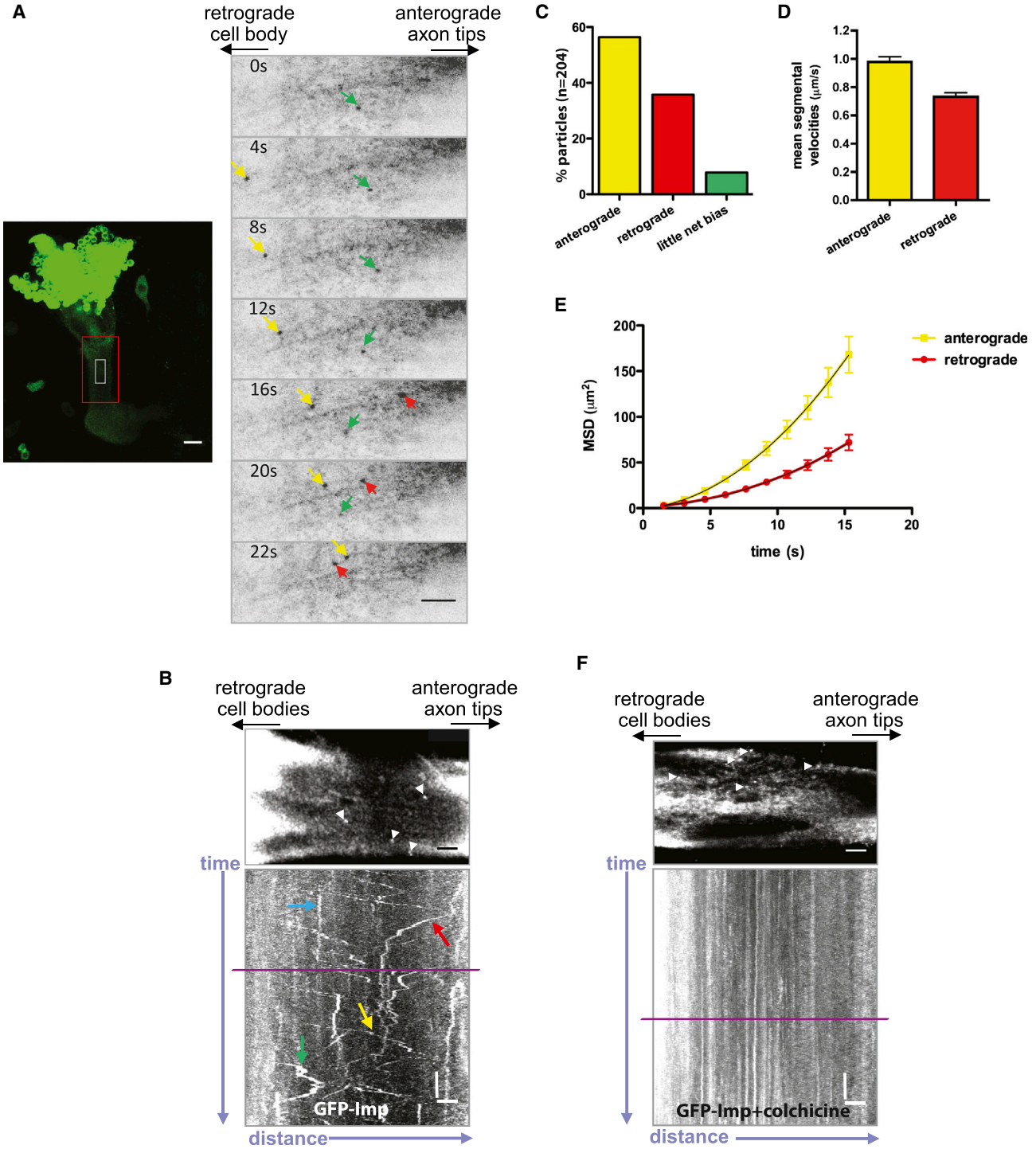



Figure 3. Imp Is Actively Transported to Growing γ Axons, in a Microtubule-Dependent Manner

(A) Left: 28 hr APF MB expressing GFP-Imp (green) in γ neurons. The red box corresponds to the region of the pedunculus imaged in the time-lapse sequences (see Movies S1 and S2) and the white box to the region highlighted in the right panel. Scale bar, 5 μm. Right: image sequence showing GFP-Imp particle movement. Particles moving in an anterograde direction (yellow arrows), moving in a retrograde direction (red arrows), or undergoing a reversal of direction (green arrows) are highlighted. Scale bar, 5μm.

(B) Single image (top) and kymograph (bottom) extracted from a time-lapse sequence. White arrowheads point to GFP-Imp particles (top). Blue, yellow, and red arrows point to stationary, anterograde-moving, and retrograde-moving particles, respectively (bottom). The green arrowhead points to a particle undergoing a reversal of direction. The purple line on the kymograph indicates the time point shown in the upper image. The genotype was 201Y-Gal4, UAS-GFP-Imp. Horizontal scale bar, 5 μm; vertical scale bar: 5 s.

(C) Relative proportions of the different classes of motile GFP-Imp particles. A total of 204 particles (115 anterograde, 73 retrograde, and 16 with little net bias), obtained from ten movies, was scored.

(D) Mean segmental velocities. Two hundred and seven anterograde and 175 retrograde segments were considered for the analysis. Error bars indicate the SEM.

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the *chic* 3′ UTR. Transcripts with the *chic* 3′ UTR localized more efficiently than transcripts lacking it (Figure 4F), suggesting that Imp binding to the 3′ UTR promotes *chic* axonal targeting. To exclude an effect of Imp on *chic* mRNA stability, we analyzed the levels of *chic* transcripts in cultured S2R$^+$ cells. No significant differences in *chic* mRNA and Chic protein levels could be observed upon *imp* inactivation in these conditions (Figures S4E and S4F).

### *chickadee* Controls Axonal Growth Downstream of *imp*

To functionally test the importance of *chic* regulation in vivo, we first analyzed the phenotypes associated with *chic* downregulation. Consistent with described roles of Profilin in regulating F-actin polymerization [24] and axonal pathfinding [25–27], we observed that *chic* mutant γ neurons fail to properly extend their axons (Figure 4H). More importantly, overexpression of *chic* significantly rescued the axonal growth defects observed in *imp* mutant neurons (Figures 4I and S4G–S4I). Similar results were obtained with two independent UAS-*chic* transgenes (Figure 4I), but not with overexpression of another regulator of F-actin polymerization (*enabled*, data not shown). These results suggest that *imp* controls axonal remodeling by regulating *chic* expression in vivo and reveal that forced accumulation of Chic protein in axons (Figures S4J and S4K) can partially compensate for the loss of *imp* function.

In conclusion, our finding that *Drosophila* Imp is required for γ axon regrowth but is dispensable for initial axonal growth suggests a novel and specific function of axonal mRNA targeting in developmental remodeling of the brain. Furthermore, these results highlight mechanistic similarities between developmental axonal regrowth and postinjury axonal regeneration, a process known to depend on axonal mRNA transport [28, 29]. Finally, our study uncovers that the translocation of Imp to γ axons is tightly linked to their developmental remodeling program. This reveals that mRNA transport machineries are subject to precise spatiotemporal regulation and may be specifically recruited in the context of developmental rewiring of the brain. It will now be interesting to identify the signals controlling the localization and the activity of mRNA transport machineries during this process.

#### Supplemental Information

Supplemental Information includes Supplemental Discussion, Supplemental Experimental Procedures, four figures, two tables, and two movies and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2014.02.038.

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(E) Graph representing the mean square displacement (MSD) values of GFP-Imp particles in function of time. MSDs are shown in yellow for anterograde (n = 82) and in red for retrograde (n = 57) particles. Curves corresponding to best-fitting quadratic functions typical of directed motion are shown in black ($R^2 > 0.99$, with $R^2$ representing the correlation coefficient; see the Supplemental Experimental Procedures). Error bars indicate the SEM.
(F) Single image (top) and kymograph (bottom) extracted from a movie generated on a 201Y-Gal4, UAS-GFP-Imp brain treated with colchicine. White arrowheads point to GFP-Imp particles (top). The purple line on the kymograph indicates the time point shown in the upper image. Horizontal scale bar, 5 μm; vertical scale bar: 5 s.
See also Figure S3 and Movies S1 and S2.

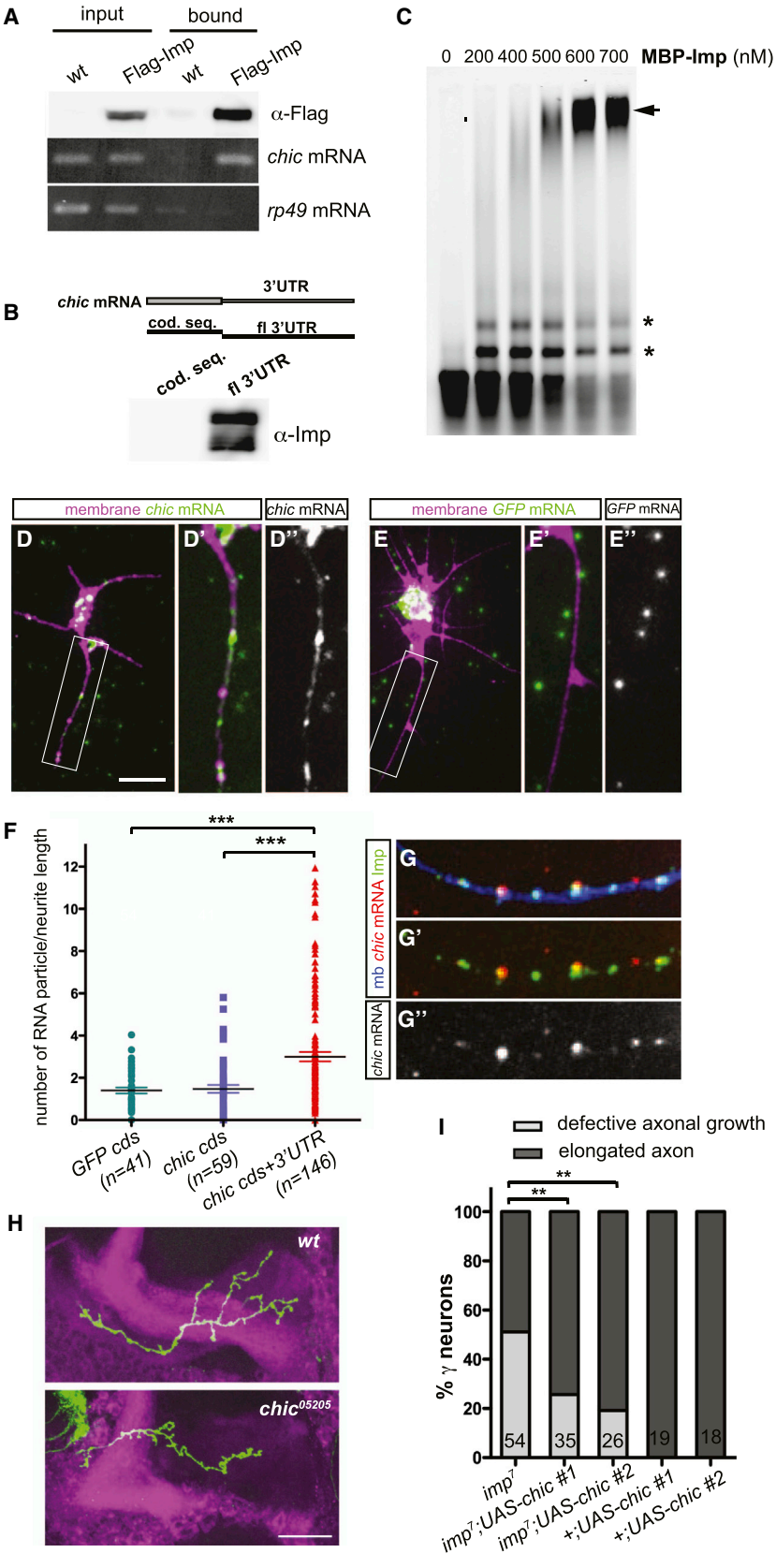



Figure 4. *chickadee* mRNA Is a Functional Target of Imp that Localizes to Developing Neurites

(A) Top: western blot performed with anti-Flag antibodies. Bottom: semiquantitative RT-PCR amplifications of mRNAs recovered in fractions immunoprecipitated from 201Y-Gal4/UAS-Flag-Imp (Flag-Imp) or 201Y-Gal4/+ (WT) brain extracts.

(B) RNA affinity pull-down assay using biotinylated RNAs incubated with wild-type brain extracts. Proteins recovered in the bound fractions were visualized after western blot analysis with anti-Imp antibodies.

(C) Electrophoretic mobility shift assays analysis using fluorescently labeled *chic* 3′ UTR in the presence of increasing amounts of MBP-Imp. The asterisks highlight low-molecular-weight complexes, and the arrow points to high-molecular-weight complexes.

(D–E″) Cultured γ neurons expressing *chic* reporter mRNAs (D; *chic* coding sequence [cds]-*chic* 3′ UTR) or a control mRNA (E; *GFP* cds) were stained with HRP to label cell membranes (magenta). Both reporters contain *SV40* 3′ UTR sequences inserted upon cloning into UASt vectors. The in situ hybridization signals obtained using *SV40* probes are shown in green in (D), (D′), (E), and (E′) and white in (D″) and (E″). The boxed areas in (D) and (E) are shown at higher magnification in (D′) and (D″) and in (E′) and (E″), respectively. Scale bar, 10 μm. Precise genotypes are as follows: 201Y-Gal4, UAS-*GFP* cds/+; UAS-Flag-*chic* cds-*chic*3′UTR/+ (D–D″) and 201Y-Gal4, UAS-*GFP* cds (E–E″). γ neurons were identified based on Flag or GFP expression (not shown).

(F) Normalized numbers of RNA particles scored in the neurites of cultured γ neurons expressing control *GFP* cds, *chic* cds, or *chic* cds-*chic* 3′ UTR reporters. Values represent the number of particles per 100 μm of neurite. n corresponds to the total number of scored neurons. ***p < 0.0001 (Kruskal-Wallis test with Dunn's post test).

(G–G″) Neurite of a cultured γ neuron expressing *chic* cds-*chic* 3′ UTR reporter RNAs and stained with HRP (blue) to label cell membranes (mb) and anti-Imp antibodies (green). The in situ hybridization signal obtained using *SV40* probes is shown in red in (G) and (G′) and in white in (G″). The precise genotype is 201Y-Gal4, UAS-*GFP* cds/+; UAS-Flag-*chic* cds-*chic*3′UTR/+.

(H) Axonal projections of single wild-type (top) and *chic*$^{05205}$ (bottom) adult γ neurons labeled by GFP (green) using the MARCM technique. The shape of MB medial lobe is highlighted by FasII staining (magenta). Scale bar, 20 μm.

(I) Percentages of adult γ axons that succeeded (elongated axon) or failed (defective axonal growth) to reach the extremity of the medial lobe. Two independent UAS-*chic* transgenes (#1 and #2) were used. Numbers correspond to the total numbers of scored axons. **p < 0.01 (Fisher's exact test).

Complete genotypes are as follows: FRT19A, tub-Gal80, hsp-flp/FRT19A *imp*$^{7}$; 201Y-Gal4, UAS-GFP/+ (*imp* mutants); FRT19A, tub-Gal80, hsp-flp/FRT19A *imp*$^{7}$; 201Y-Gal4, UAS-GFP/+; UAS-*chic*/+ (rescue condition); FRT19A, tub-Gal80, hsp-flp/FRT19A; 201Y-Gal4, UAS-GFP/+; UAS-*chic*/+ (control condition). See also Figure S4 and Table S2.

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